



MERIDIAN JUNIOR COLLEGE
JC2 Preliminary Examination
Higher 2

H2 Mathematics

9758/02

Paper 2

19 September 2018

3 Hours

Additional Materials: Writing paper

Graph Paper

List of Formulae (MF 26)

READ THESE INSTRUCTIONS FIRST

Write your name and civics group on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use a graphing calculator.

Unsupported answers from a graphing calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphing calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

Section A: Pure Mathematics [40 marks]

- 1 (a)** Consider the polynomial $P(z) = a_0 + a_1z + a_2z^2$, where $a_0, a_1, a_2 \in \mathbb{C}$, $a_2 \neq 0$. Show that if the equation $P(z) = 0$ has a complex root w , then its complex conjugate w^* must also be a root to the equation. [3]

- (b)** The polynomial $P(z)$ has real coefficients. The equation $P(z) = 0$ has a root $z_1 = -a + ai$, where a is a positive real number.

(i) Write down a second root, z_2 , in terms of a . [1]

(ii) Hence sketch the points Z_1 and Z_2 , representing z_1 and z_2 respectively, on an Argand diagram. [2]

(iii) Deduce the area of OZ_1Z_2 , in terms of a . [1]

- 2 (i)** Using the formulae for $\cos(A \pm B)$, prove that

$$\cos\left(r + \frac{1}{2}\right)\theta - \cos\left(r - \frac{1}{2}\right)\theta \equiv -2\sin r\theta \sin \frac{1}{2}\theta. \quad [2]$$

(ii) Hence show that $\sum_{r=1}^n \sin r\theta = -\frac{1}{2} \operatorname{cosec} \frac{1}{2}\theta \left[\cos\left(n + \frac{1}{2}\right)\theta - \cos \frac{1}{2}\theta \right]$ [3]

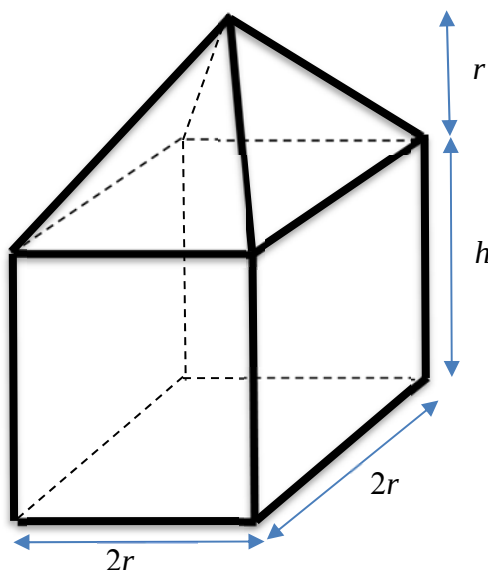
(iii) Using the result in **(ii)**, find the exact value for $\sum_{r=5}^{62} \sin(r-2)\theta$ when $\theta = \frac{\pi}{3}$. [4]

- 3 (a) A semicircle has radius r cm, perimeter P cm and area A cm². Show that

$$\frac{dP}{dA} = \frac{2 + \pi}{\pi r}.$$

Determine the exact value of the radius when the area of the semicircle is increasing at a constant rate of 3 cm²/s and the perimeter is increasing at a constant rate of $\frac{3}{5}$ cm/s. [4]

- (b) An architectural firm wants to make a model of a greenhouse as shown.



The model is to be made up of three parts.

- The roof is modelled by a pyramid with a square base $2r$ cm by $2r$ cm and whose apex is r cm directly above the center of its base.
- The walls are modelled by four rectangles measuring $2r$ cm by h cm.
- The floor is modelled by a square measuring $2r$ cm by $2r$ cm.

The three parts are joined together as shown in the diagram. The model is made of material of negligible thickness.

It is given that the external surface area of the model has a fixed value of 160 cm². Show that $V = 80r - \left(\frac{2}{3} + 2\sqrt{2}\right)r^3$. Hence, using differentiation, find the value of r which gives the maximum value of V . (You do not need to verify that the volume is a maximum for this value of r .) [Volume of Pyramid = $\frac{1}{3} \times \text{base area} \times \text{height}$] [7]

- 4 (a) (i)** Given that f is a continuous function, explain, with the aid of a sketch, why the value of

$$\lim_{n \rightarrow \infty} \frac{1}{n} \left\{ f\left(\frac{1}{n}\right) + f\left(\frac{2}{n}\right) + \dots + f\left(\frac{n}{n}\right) \right\}$$

is $\int_0^1 f(x) dx$. [2]

- (ii)** Hence evaluate $\lim_{n \rightarrow \infty} \frac{1}{n} \left\{ e^{\frac{3}{n}} + e^{\frac{6}{n}} + \dots + e^3 \right\}$, leaving your answer in exact form. [3]

- (b)** The function f is defined by

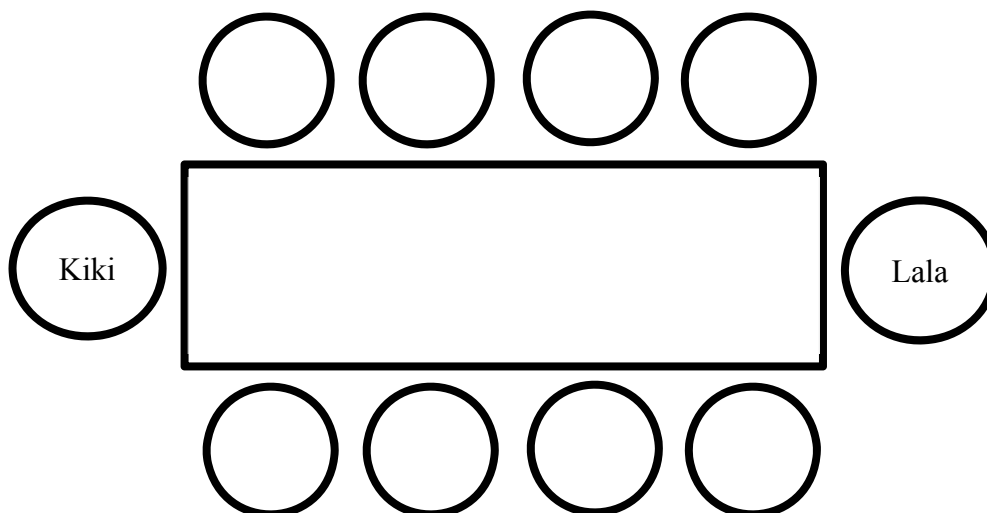
$$f(x) = \begin{cases} (x+1)^2 + a & \text{for } -1 < x \leq 1, \\ (4+a)(2-x) & \text{for } 1 < x \leq 2, \end{cases}$$

where a is a positive real constant and that $f(x) = f(x+3)$ for all real values of x .

- (i)** Evaluate $f(-41)$ and $f(2018)$. [2]
- (ii)** Sketch the graph of $y = f(x)$ for $-4 < x \leq 6$. [3]
- (iii)** Hence find the value of $\int_{-4}^4 f(x) dx$ in terms of a . [3]

Section B: Statistics [60 marks]

- 5 Kiki and Lala host a dinner for four other married couples. They sit at a rectangular table with Kiki and Lala at the left and right ends of the table respectively as shown in the diagram below.



Find the number of ways to seat the four couples such that

- (i) there are no restrictions. [1]
- (ii) each married couple is seated directly facing each other on opposite sides. [2]
- (iii) each married couple is seated directly facing each other on opposite sides and two particular ladies cannot be seated next to each other on the same side. [3]

- 6** A bag contains five balls numbered 1 to 5. A game consists of a player taking two balls from the bag consecutively, with replacement. The number on each ball is noted down. His score X is found by taking the absolute value of the difference between the two numbers.

(i) Obtain the probability distribution of X . [2]

(ii) Find $E(X)$ and $\text{Var}(X)$. [2]

A player pays \$ a to play this game. He wins an amount (in dollars) corresponding to twice his score. Determine the range of values of a if the player is expected to make a loss. [2]

- 7** During the National Day Rally in 2017, PM Lee cited that an average of one in nine Singaporeans has diabetes. Suppose we examine 24 patients who are randomly selected from a polyclinic. Let X be the number of these patients who have diabetes.

(i) State, in the context of this question, two assumptions needed to model X by a binomial distribution. [2]

(ii) Explain why one of the assumptions stated in part **(i)** may not hold in this context. [1]

Assume now that these assumptions do in fact hold.

(iii) Find the probability that in a sample of 25 randomly selected patients, the 25th patient is the third patient who has diabetes. [2]

(iv) Find the least value of n such that the probability of having at most n diabetic patients in the random sample of 24 patients is greater than 0.9. [3]

- 8** A circular disc is divided into ten equal sectors where five sectors are coloured blue, three sectors are coloured green and two sectors are coloured red.

In a game, a player is given a maximum of three times to spin a pointer pivoted at the centre of the disc.

- If the pointer lands on a blue sector, the player wins \$2 and the game continues.
 - If the pointer lands on a green sector, the game ends and the player loses all of his winnings.
 - If the pointer lands on a red sector in the first spin, the game ends and the player wins the grand prize of \$20.
 - If the pointer lands on a red sector in the second or third spin, the player wins double the amount of his total winnings from all his previous spins, and the game continues. For example, if the pointer lands on a blue sector in the first spin and on a red sector in the second spin, the total winnings for the first two spins will be \$6.
- (i)** Construct a probability tree showing this information. [3]
- (ii)** Find the exact probability that the player wins \$12 or more when the game ends. [2]
- (iii)** Find the exact probability that the pointer in the second spin lands on a blue sector given that the player wins \$12 or more when the game ends. [2]
- (iv)** Suppose there is no limit to the number of times a player can spin, find the exact probability that the player has no spins in which the pointer lands on a red sector and wins nothing when the game ends. [2]

- 9** Based on past records, shoppers spend an average of 1.5 hours at MJ mall per visit. After a revamp and expansion to include more stores, the mall management wishes to conduct a hypothesis test to check if the duration that shoppers spend in the mall per visit has increased. A random sample of 50 shoppers were surveyed and the duration of time, x , in hours per visit is recorded. The sample sum is 82.5 hours and the sample variance is 0.425 hours².
- (i)** State appropriate hypotheses for the test and find the unbiased estimates of the population mean and variance. [3]
 - (ii)** Explain why the mall management is able to carry out a hypothesis test without making any assumptions about the distribution of the duration shoppers spend at MJ mall per visit. [1]
 - (iii)** Calculate the p -value and state its meaning in context of the question. Hence state the set of values of the level of significance $\alpha\%$ for which the management's hypothesis is valid. [4]
 - (iv)** The mall management believes that the mean spending per shopper per visit is \$150. A test at 5% level of significance found that there is significant evidence that the population mean spending is less than \$150. Using only this information, explain if the following statements are necessarily true, necessarily false, or neither necessarily true nor necessarily false.
 - (a)** There is significant evidence at the 10% level of significance that the population mean spending is less than \$150.
 - (b)** There is significant evidence at the 5% level of significance that the population mean spending differs from \$150. [2]

- 10** Biologists monitor the population of wild rabbits in a dry grassland region of the Australian outback. The population of wild rabbits, y , in hundreds, in month x are as follows.

Month x	1	3	6	9	11	13	15
Population (in hundreds) y	40.3	35.2	36.3	40.8	39.7	41.5	42.1

- (i) Draw a scatter diagram showing these data. [1]
- (ii) Suggest a possible reason, in context, why one of the data points does not seem to follow the trend. [1]
- (iii) It is desired to predict the population of rabbits in the future months. Explain why, in this context, a linear model is not appropriate. [1]

After removing the outlier, the biologists decided to fit a model of the form $\ln(M - y) = a + bx$, where M is a suitable constant for the remaining data points. The product moment correlation coefficient between x and $\ln(M - y)$ is denoted by r . The following table gives values of r for some possible values of M .

M	44	45	46
r	- 0.945 943		- 0.945 994

- (iv) Calculate the value of r for $M = 45$, giving your answer correct to 6 decimal places. [1]
- (v) Use the table and your answer in part (iv) to suggest with a reason which of 44, 45 or 46 is the most appropriate value for M . [1]
- (vi) Using the value for M , calculate the values of a and b , and use them to predict the population of the wild rabbits after 2 years, to the nearest whole number. [4]
- (vii) Give an interpretation, in context, of the value of M . [1]

- 11 (a)** The time T (in minutes) taken by Kathy to drive from her house to the nearest supermarket has a mean of 5 and a variance of 8.

Explain why T is unlikely to be normally distributed. [2]

- (b)** In this question you should state clearly the values of the parameters of any normal distribution you use.

In a supermarket, the masses in kilograms of chickens have the distribution $N(2.4, 0.5^2)$ and the masses in kilograms of ducks have the distribution $N(4.3, 1.8^2)$.

- (i)** Find the probability that the total mass of 2 randomly chosen chickens is more than 5kg. [2]

- (ii)** Find the probability that the mean mass of a randomly chosen duck and 2 randomly chosen chickens is more than 3.20kg. [3]

Chickens are sold at \$7 per kilogram and ducks are sold at \$13 per kilogram. The supermarket is conducting their annual sale and has a discount of 10% and 25% off the prices of chickens and ducks respectively.

- (iii)** Find the probability that the total cost of 2 randomly chosen chickens and a randomly chosen duck is less than \$80. [4]

End of Paper